Machine Learning and Data Mining Project:
Food Tweet Popularity Prediction

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1. Problem Statement

The rise of social networks has generated a large amount of textual data making it possible to analyze online human interactions. In this project, we focus on food-related tweets in order to identify the factors influencing their popularity. Unlike a simplistic approach, the number of likes alone does not capture the full engagement of a tweet. Indeed, several forms of interaction exist, including likes, retweets, replies, quotes, and, to some extent, implicit reactions such as emojis present in the textual content. Thus, a more complete measure of popularity is necessary. We therefore define popularity as a composite variable integrating both engagement metrics and sentiment: Popularity = Likes + Retweets + Replies + Quotes + Sentiment. This definition makes it possible to capture the full set of social interactions while taking into account the emotional dimension of the content, and to better model content virality. Moreover, the sentiment expressed in tweets (positive, negative or neutral), computed using the TextBlob library, is also an important factor that may influence popularity, since emotional content tends to generate more engagement.

The main objective is therefore to model the popularity of food-related tweets, to identify the most influential variables, especially textual features, user information, and sentiment, and to propose a robust predictive model capable of capturing complex relationships between these different variables.

This is a supervised learning regression problem, in which models are trained on labeled data (tweets) in order to predict a continuous target variable corresponding to tweet popularity.

2. Algorithms used to Approach the Problem

Sentiment and emotion analysis related to a tweet is performed using the TextBlob library (a sentiment classification model based on Python; built in 2013 [2]) in order to determine whether a tweet expresses a positive, negative, or neutral sentiment by assigning a polarity score ranging from -1 (negative) to +1 (positive). The TF-IDF (Term Frequency – Inverse Document Frequency) method was used to transform textual data into numerical vectors, allowing identification of the most representative words in the content. For predictive modeling, two approaches were compared. Linear regression serves as a baseline model by capturing simple relationships between variables, but it remains limited in the presence of non-linear relationships. To better represent these interactions, the Random Forest Regressor model [1] was used, offering robustness and the ability to handle complex and noisy data.

3. Data description

The Food_Tweets dataset [3] contains a total of 50 000 food-related tweets. It includes a total of 22 variables, including: tweet.id, tweet.text, tweet.lang, tweet.datetime, tweet.public_metrics.retweet_count, tweet.public_metrics.reply_count, tweet.public_metrics.like_count, tweet.public_metrics.quote_count, user.username, user.fullname, user.description, user.location, user.public_metrics.followers_count, user.public_metrics.following_count, user.public_metrics.tweet_count, user.public_metrics.listed_count, media.type, media.width, media.height, media.duration_ms, media.url and media.public_metrics.view_count. The tweets are mostly written in English and cover a variety of contexts, including personal experiences, festive events, promotional content, and social discussions related to food. The analysis of extreme values shows that the tweet with the highest number of likes (37 992) and retweets (5 813) was published by the user "HowThingsWork_". The highest number of replies (2 505) is associated with "AZ_Brittney", while the highest number of quotes (1 058) comes from "SnipedPaulie". Regarding account influence, the user "CNN" has the highest number of followers with 60 873 705 subscribers. Moreover, the content that generated the highest number of views (14 019 735) is associated with the user "tracedontmiss". Finally, we observe a highly imbalanced distribution of the data, where most tweets show a low level of interaction, while a limited number of tweets reach very high popularity.

4. Experimental Procedure

For the experimental procedure, the dataset was restricted to tweets written in English in order to ensure consistency in text processing and sentiment analysis. Observations with missing values or incomplete information, especially in engagement metrics or user features, were removed, as they did not provide relevant information for modeling.

4.1. Data cleaning and Preprocessing

The following steps were considered: removal of punctuation and numerical characters, conversion of all letters to lowercase, elimination of unnecessary whitespace as well as irrelevant terms, removal of stopwords and low-information words, and overall text cleaning in order to retain only useful information. A sentiment metric was then computed from the tweet textual content. Finally, texts were transformed into a TF-IDF vector representation, which automatically performs tokenization and converts tweets into a matrix usable for prediction models.

4.2. Results and Discussion

The obtained results are based both on exploratory data analysis and on the performance of predictive models.

4.2.1 Exploratory data analysis

Figure 1 shows the distribution of likes. It is highly skewed, with a large majority of tweets receiving a low number of interactions, while a very small number of tweets reach extremely high values.

Figure 2 illustrates the relationship between the number of followers and tweet popularity.

Figure 3 presents the distribution of sentiment scores.

4.2.2 Predictive model results

Two models were evaluated: linear regression and Random Forest Regressor.

Linear regression shows limited performance. The Random Forest model shows better performance due to non-linear modeling capacity.

4.2.3 Discussion

Tweet popularity cannot be explained by a single variable. Textual content and sentiment are key factors. Data imbalance remains a major challenge.

4.2.4 Conclusion of results

Popularity depends on multiple factors including content, sentiment, and user influence. Random Forest is more effective than linear regression.

References

[1] L. Breiman, Random Forests, Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.
[2] S. Loria, TextBlob: Simplified Text Processing, 2013. [Online]. Available: https://textblob.readthedocs.io.
[3] tweets.csv Dataset, dataset containing 50 000 tweets with engagement metrics and user information, 2022.

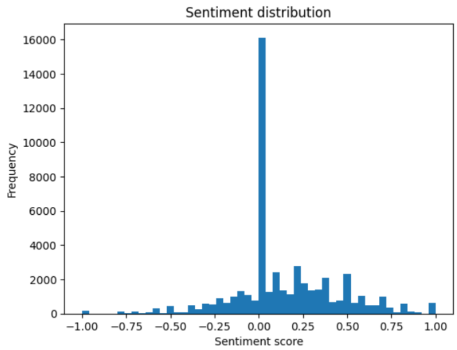

Figure 1: Likes distribution

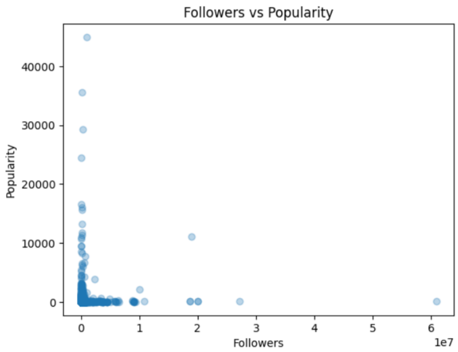

Figure 2: Followers vs Popularity

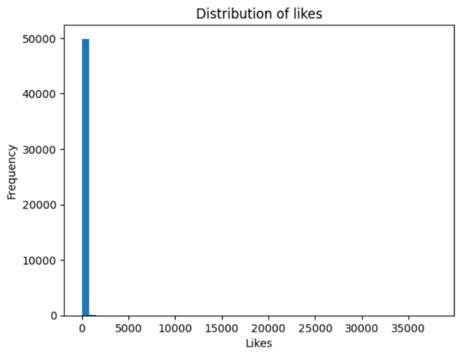

Figure 3: Sentiment distribution